ثانوية المدينة التعليمية
Education City High School

1.5

Polynomial Functions and Complex Zeros

Student Notes and Guided Practice

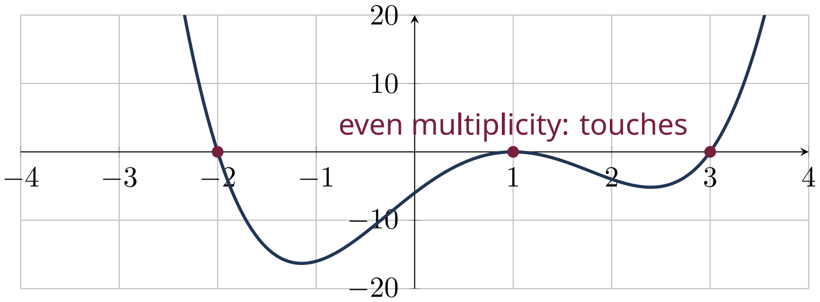
AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Use the Fundamental Theorem of Algebra to count complex zeros with multiplicity.
- Apply the complex conjugate zero theorem for real-coefficient polynomials.
- Connect zeros and multiplicities to intercept behavior.

Essential Question

How do algebraic factors and complex zeros determine a polynomial graph?



Concept Framework

Key Idea

A degree-n polynomial has exactly n complex zeros counting multiplicity.

Key Idea

Nonreal zeros of a real-coefficient polynomial occur in conjugate pairs.

Key Idea

Odd multiplicity usually produces crossing at a real zero; even multiplicity produces touching.

Key Idea

A zero r corresponds to a factor $x - r$.

Formula Map

Formula and Structure

$$P(r) = 0 \iff (x - r) \text{ is a factor}$$

Formula and Structure

$$z = a + bi \Rightarrow \bar{z} = a - bi \text{ is also a zero}$$

Formula and Structure

$$\sum \text{multiplicities} = \deg P$$

Worked Examples

Worked Example 1

Problem. A real quartic has zeros 2 and $1 + 3i$. List all zeros if 2 has multiplicity 2.
Solution. The conjugate $1 - 3i$ is also a zero. The four zeros counting multiplicity are $2, 2, 1 + 3i, 1 - 3i$.

Worked Example 2

Problem. Construct a monic real cubic with zeros -1 and $2 + i$.
Solution. Include $2 - i$. Then $(x + 1)[(x - 2)^2 + 1] = (x + 1)(x^2 - 4x + 5)$.

Worked Example 3

Problem. At $x = 3$, a graph flattens and crosses. What multiplicity is plausible?
Solution. An odd multiplicity greater than 1, commonly multiplicity 3.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Analyze zeros and multiplicities for $P(x) = (x+2)^2(x-1)^3$.

2. Analyze zeros and multiplicities for $P(x) = x^4 - 5x^2 + 4$.

3. Analyze zeros and multiplicities for $P(x) = x^3 + 4x$.

4. Analyze zeros and multiplicities for $P(x) = x^4 + 4$.

5. Analyze zeros and multiplicities for $P(x) = (x-3)^4(x+1)$.

6. A real polynomial of degree 6 has zeros 2 (multiplicity 2), -1, and $3+4i$. What zeros are still required and how many degrees remain?

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is ____________________.
2. A restriction I must remember is ____________________.
3. One graphical feature I can justify algebraically is ____________________.

Video support: Flipped Math topic page | Original ECHS material informed by Pearson and Holt Precalculus concepts.